Constrained-worldline brachistochrones in non-stationary spacetimes: Kodama rails, closed forms, and plunge inversion

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We extend the relativistic brachistochrone (fastest-descent constrained rail curves, Perlick 1991) from stationary to non-stationary spacetimes. The rail constraint $-u \cdot W = \hat{E}$ is an actively controlled invariant, legitimate via Pontryagin's principle; the selector W follows a hierarchy Killing $\to$ conformal Killing $\to$ Kodama vector. We treat FLRW (degenerate base), Vaidya (dynamical black hole, $W = \partial_v$ the Kodama vector), and Thakurta–Kerr (conformal rotating black hole). Main results: Kodama energy conservation along the rail; the plunge-radius law and the absence of a *physical* evaporative inversion (only rotation and the conformal factor invert); closed-form equatorial separatrices in Weierstrass functions with Doran continuation through the ergosphere; the trichotomy at the ergosphere with an analytic cusp theorem; and an existence theorem for the conformal plunge inversion.

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I. INTRODUCTION

The relativistic brachistochrone—the fastest constrained "rail" worldline between two events—was formulated by Perlick for stationary spacetimes [1, 2], where a timelike Killing vector provides the conserved rail energy. It connects naturally to Fermat's principle and to optical (Randers–Zermelo–Finsler) geometry [3–7]. Building on our earlier relativistic-brachistochrone studies—the stationary Kerr geometry [8], the interior Schwarzschild metric [9], and Tolman–Oppenheimer–Volkoff stellar interiors [10]—we extend the construction to *non-stationary* spacetimes: FLRW, Vaidya [11], and Thakurta–Kerr [12]. These extensions raise three questions that organize the paper. *(i) Is the rail constraint legitimate without a Killing vector?* We show it is: $-u \cdot W = \hat{E}$ is an actively controlled invariant, and the fastest constrained worldline solves a well-posed time-optimal control problem (Sec. II) whose control cost measures precisely the missing symmetry. *(ii) What replaces the conserved energy?* In a spherically symmetric dynamical spacetime the selector W is the Kodama vector, whose flux charge is the Misner–Sharp mass; the rail conserves the Kodama energy identically even though it is *not* Noether-conserved for free fall (Sec. IV). *(iii) Which phenomena are genuinely non-stationary?* We isolate three: capture by cosmological expansion, the teleological (future-anticipating) turning point in Vaidya, and the breathing freezing surface that eats the radius of convergence of the quasi-constants in Thakurta–Kerr. The three spacetimes form a ladder of increasing structure—FLRW (degenerate base), Vaidya (dynamical horizon), Thakurta–Kerr (rotation + conformal expansion)—treated in Secs. III–V; equatorial closed forms and the plunge-inversion analysis follow in Secs. VI–VI D.

II. THE CONTROLLED-RAIL PRINCIPLE

A. Rail constraint and the indicatrix

The rail is a constrained worldline with the invariant

$$-\,u\cdot W \;=\; \hat{E}, \tag{1}$$

actively maintained by an ideal (magnetic-type) rail force $f_\mu = Du_\mu/\mathrm{d}\tau$ with $f\cdot u = 0$, so that the rail does no work on the particle.

a. The ellipse. Fix an event and a time function adapted to W. The linear constraint (1) fixes the W-component of u; the mass shell $u\cdot u = -1$ is a quadric; their intersection, projected onto the physical velocity plane $(\mathrm{d}r,\, r\,\mathrm{d}\varphi)$ per unit selector-time, is an ellipse

$$\dot{x}(\theta) = c + R(\theta), \qquad \theta \in [0, 2\pi), \tag{2}$$

whose center c is the Zermelo wind (zero for FLRW, ingoing radial for Vaidya, azimuthal—frame dragging—for Thakurta–Kerr) and whose axes R shrink with the local speed as the freezing surface is approached (Fig. 1). Every branch—arrival time t, proper time τ, conformal time η—shares this *same* indicatrix; the branches differ only in the cost one-form, i.e. in which clock measures the elapsed travel time.

b. Pontryagin reduction. Minimizing travel time subject to (2) is a time-optimal control problem with control θ. Since the admissible-velocity set is a smooth strictly convex oval, the maximizer of $p\cdot\dot{x}(\theta)$ is unique and Pontryagin's maximum principle [13, 14] delivers the branch Hamiltonian $H(x, p)$ in closed form (Secs. IV–V); the convex-oval regularity makes the vakonomic and d'Alembert formulations coincide, so no Filippov relaxation is needed. Two structural facts hold throughout: the transversality condition

$$H = 0 \qquad \text{(free arrival time)}, \tag{3}$$

and $p_\varphi = J$ conserved (axial symmetry). In the stationary limit H is autonomous and $H \equiv 0$ is a genuine first integral; non-stationarity enters only through the explicit time dependence $\mathrm{d}H/\mathrm{d}(\text{time}) = \partial_{\text{time}} H$, proportional to $m'(v)$ (Vaidya) or $A'(\eta)$ (Thakurta–Kerr).

c. Cost of control. Differentiating the invariant along the worldline,

$$\frac{\mathrm{d}(-u\cdot W)}{\mathrm{d}\tau} = -\,u^a u^b\,\nabla_{(a}W_{b)}, \tag{4}$$

which vanishes for *all* u iff W satisfies the Killing equation. When it does not, the right-hand side is the power the rail force must supply to hold $\hat{E}$ fixed: the control cost equals the failure of the Killing equation, i.e. the missing symmetry. This is why the non-stationary rail is legitimate but not free—$\hat{E}$ is controlled, not conserved. The optical reductions of H reproduce the Randers/Zermelo structure [6, 15].

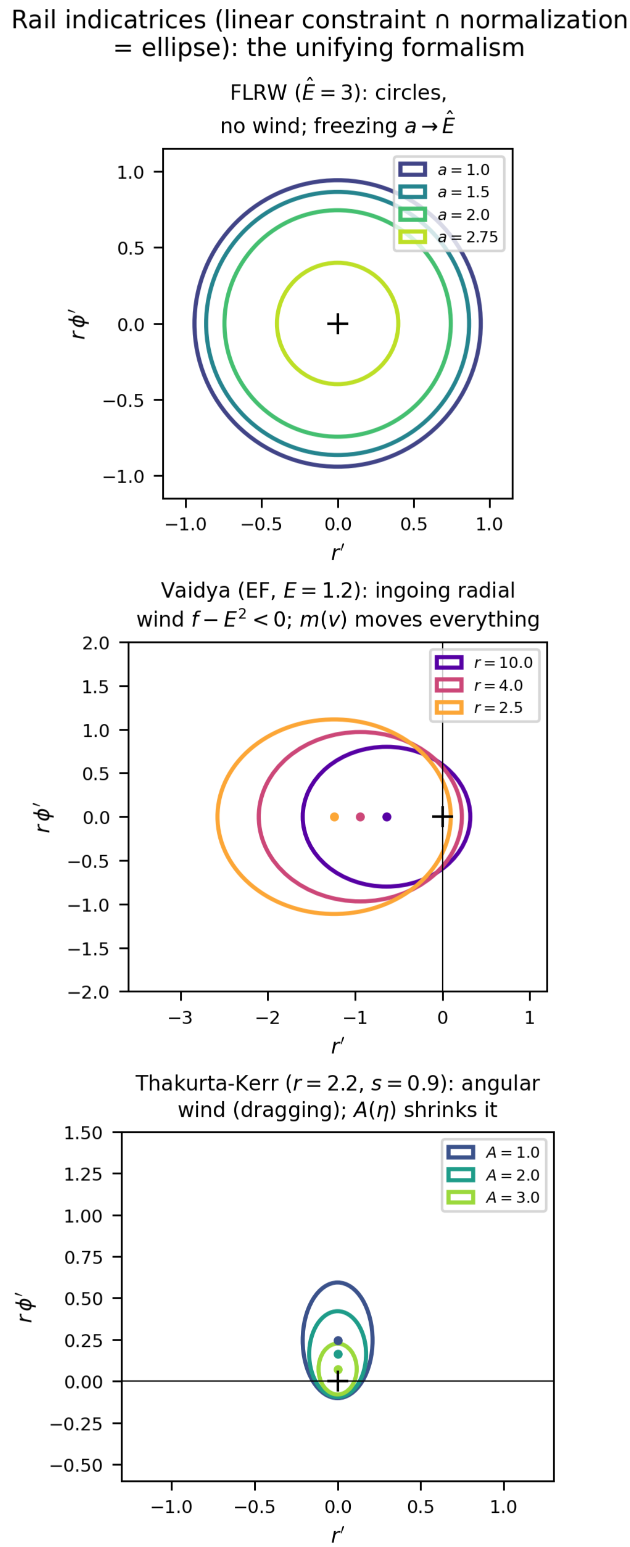

FIG. 1. Rail indicatrices (linear constraint ∩ normalization = ellipse): the unifying formalism. FLRW gives shrinking circles (no wind); Vaidya an ingoing radial wind ($f - E^2 < 0$, driven by $m(v)$); Thakurta–Kerr an angular wind (frame dragging), the ellipse shrinking with $A(\eta)$.

B. Selecting W: the Killing–CKV–Kodama hierarchy

The selector W is fixed by a hierarchy of decreasing symmetry, each level recovering the previous one in the appropriate limit.

(a) Killing. If a timelike Killing vector ξ exists, $W = \xi$ and the right-hand side of Eq. (4) vanishes identically: $-u \cdot \xi$ is Noether-conserved and the rail is free. This is Perlick's stationary case [1].

(b) Conformal Killing. If the spacetime is conformally stationary, $g = \Omega^2 \hat{g}$ with $\hat{g}$ stationary, the timelike conformal Killing vector ∂_η selects the conformal-time branch; $-u \cdot W$ is exactly conserved for null rays and controlled up to the conformal factor for massive rails. FLRW and constant-A Thakurta–Kerr live at this level.

(c) Kodama. In a spherically symmetric *dynamical* spacetime with no Killing or conformal Killing vector, the Kodama vector $K^a = \varepsilon^{ab}\nabla_b r$ [16, 17] is the canonical replacement: it is divergence-free (Eq. (8)), reduces to ξ in the static limit, and its flux $G^{ab}K_b$ is conserved with charge the Misner–Sharp mass. Vaidya is the paradigm (Sec. IV A).

(d) Convention. Absent even spherical symmetry the Kodama construction fails and W becomes a declared convention, with no distinguished normalization of the cost one-form. Levels (a)–(c) are what make the non-stationary rail canonical rather than arbitrary.

III. FLRW: THE DEGENERATE BASE CASE

For spatially flat FLRW, $g = a(\eta)^2 \eta_{\mathrm{Mink}}$, the rail uses the conformal Killing vector ∂_η; the local speed is

$$v(\eta) = \sqrt{1 - a(\eta)^2/\hat{E}^2}, \tag{5}$$

vanishing at the freezing barrier $a = \hat{E}$.

a. Branch degeneracy. Because $a = a(\eta)$ depends on conformal time alone (spatial homogeneity), the optical index built from the conformal Killing vector,

$$n(\eta, r) = \frac{1}{v} = \frac{\hat{E}}{\sqrt{\hat{E}^2 - a(\eta)^2}}, \tag{6}$$

is *position-independent* at fixed η. The conformal-time functional $T_\eta = \int n \, \mathrm{d}\ell_{\mathrm{flat}}$ is then minimized by a flat straight line (Fermat with a purely time-dependent index). The same curve minimizes T_t and T_τ: with $\mathrm{d}t = a\,\mathrm{d}\eta$ and $\mathrm{d}\tau = a\sqrt{1 - \dot{\vec{x}}^2}\,\mathrm{d}\eta$, the monotone reparametrization $t(\eta) = \int a\,\mathrm{d}\eta$ maps critical points to critical points, and the τ-cost differs from the η-cost only by the same η-dependent weight. Hence

$$\begin{aligned} \arg\min T_t = \arg\min T_\tau &= \arg\min T_\eta \\ &= \text{comoving straight line.} \end{aligned} \tag{7}$$

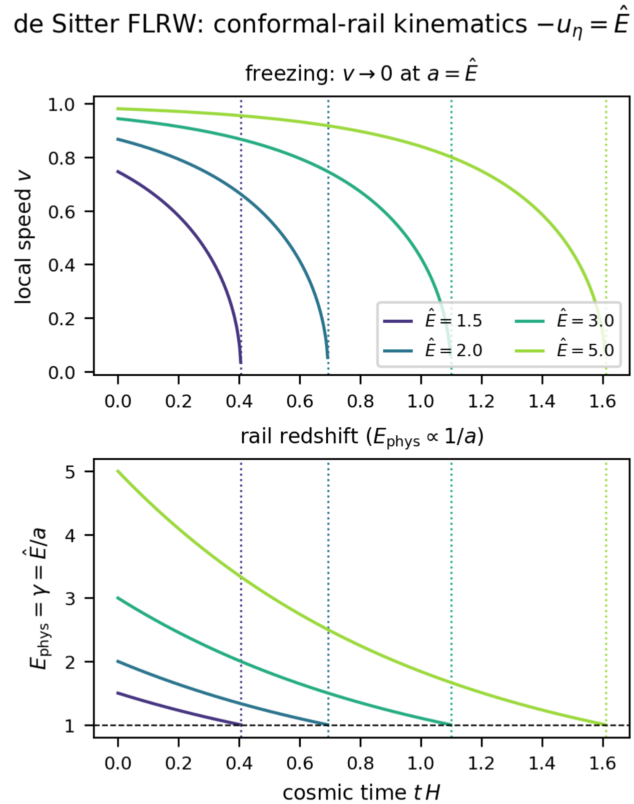

FIG. 2. de Sitter FLRW rail kinematics: local speed v and rail redshift $E_{\mathrm{phys}} = \hat{E}/a$; freezing at $a = \hat{E}$.

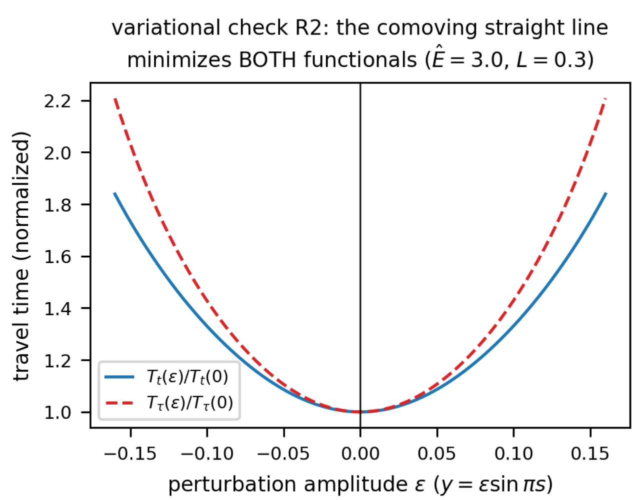

FIG. 3. Variational check: the comoving straight line minimizes *both* T_t and T_τ (branch degeneracy).

The t/τ splitting that drives every later result is a gravitational-*gradient* effect ($\partial_r f \neq 0$); it is absent whenever the metric functions depend on time only. FLRW is thus the degenerate base of the ladder: it exhibits freezing at $a = \hat{E}$ (Fig. 2) but no branch structure (Fig. 3).

IV. VAIDYA: DYNAMICAL BLACK HOLE AND THE KODAMA RAIL

The ingoing Vaidya metric $\mathrm{d}s^2 = -f\,\mathrm{d}v^2 + 2\,\mathrm{d}v\,\mathrm{d}r + r^2\mathrm{d}\Omega^2$, $f = 1 - 2m(v)/r$ [11, 18], has no timelike Killing vector; its trapping structure is that of a dynamical horizon [19–21]. The Kodama vector [16, 17, 22]

$$K^a = \varepsilon^{ab}\nabla_b r = \partial_v, \qquad \nabla_a K^a = 0, \tag{8}$$

is divergence-free without any Killing symmetry, and its charge is the Misner–Sharp mass $m(v)$ [23, 24].

A. Kodama energy is the rail invariant

From the rail constraint alone one finds the exact identity $-g(T,T) = (f\,u^v - 1)^2/\hat{E}^2$, hence

$$-\,u\cdot K \;=\; \hat{E} \quad \text{identically along the brachistochrone.} \tag{9}$$

This is *controlled* conservation, not Noether: for a geodesic $-u\cdot K$ would drift at rate $-m'(u^v)^2/r \propto m'$, exactly the right-hand side of Eq. (4) for $W = K$. Only the unnormalized $K = \partial_v$ has zero control cost in the static limit; the normalized $\hat{K} = K/\sqrt{f}$ does not (discriminating test), which is why the Kodama vector, not a unit selector, is the canonical choice.

a. Indicatrix and Hamiltonians. With $w \equiv \hat{E}^2 - f$ the constraint $-u_v = \hat{E}$ turns the admissible velocities at fixed (v,r) into an ellipse,

$$r' = (f - \hat{E}^2) + \hat{E}\sqrt{w}\,\cos\theta, \qquad \varphi' = \frac{\sqrt{w}}{r}\sin\theta, \tag{10}$$

a Zermelo problem with ingoing radial wind $f - \hat{E}^2 < 0$. Maximizing $p\cdot\dot{x}$ over θ at fixed $p_\varphi = J$ gives the closed-form branch Hamiltonians

$$H_v = p_r(f - \hat{E}^2) - 1 + \sqrt{w}\,\sqrt{\hat{E}^2 p_r^2 + J^2/r^2}, \tag{11}$$

$$H_\tau = p_r(f - \hat{E}^2) - \hat{E} + \sqrt{w}\,\sqrt{(\hat{E}p_r + 1)^2 + J^2/r^2}. \tag{12}$$

The τ-branch is the v-branch under $p_r \to p_r + 1/\hat{E}$ with unit cost replaced by $\hat{E}$. Non-stationarity is the explicit dependence $\mathrm{d}H/\mathrm{d}v = \partial_v H \propto m'(v)$; the teleological momentum $p_v = -H$ satisfies $p_v(r_1) = 0$ at the free arrival radius yet $p_v(r) \neq 0$ along the path, so the turning point *anticipates future accretion* (Fig. 5).

B. Phenomenology: penetration, timing, bounce

Three features distinguish the dynamical hole from its frozen-mass Schwarzschild limit.

Penetration. The τ-brachistochrone reaches the apparent horizon $r = 2m(v)$ only if $|J|$ lies below a threshold $J_c(v_0)$; accretion ($m' > 0$) opens a *finite* window $J_c > 0$, whereas evaporation ($m' < 0$) closes it to zero measure—an accretion/evaporation asymmetry with no static analogue. The finite capture interval $J \in (0, J_c]$ is the analogue of the Kerr t-branch dichotomy, generated here by the *dynamics*: grazing orbits linger near periapsis while the growing horizon closes the gap.

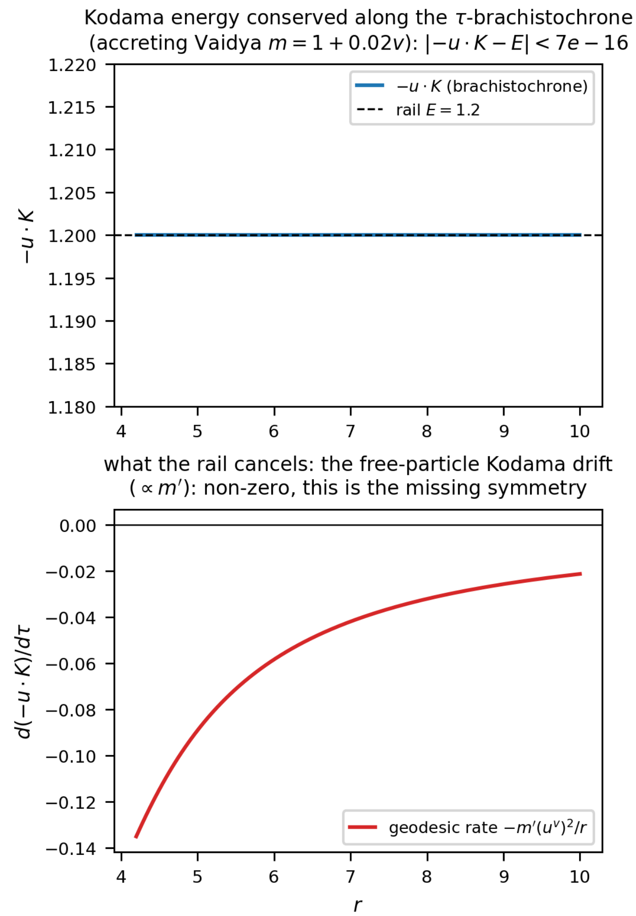

FIG. 4. Kodama energy $-u\cdot K = \hat{E}$ conserved along the τ-brachistochrone to 6.7×10^{-16} (top); the free-particle drift $\propto m'$ that the rail cancels (bottom).

Bounce. The r-parametrization degenerates at periapsis ($\mathrm{d}v/\mathrm{d}r \to \infty$); reparametrizing by v makes the Euler–Lagrange problem regular at the turning point $r' = 0$, so the trajectory is integrated straight *through* the bounce—the same extremal, via the exact change of variable $(fv' - 1)\mathrm{d}r = (f - r')\mathrm{d}v$. The v-flow reaches the periapsis exactly where the r-flow diverges ($r_{\min} = 3.0105$ at $v = 7.29$ for $\mu \equiv m' = 0.01$), and the teleological memory $p_v(r) \propto m'$ measures the departure from the frozen-mass Schwarzschild curve (Fig. 20).

Timing and the orbit–horizon race. Because m grows along the path, $r_{\min}$ depends on *when* the particle arrives, the epoch v_1—a genuinely non-stationary effect absent in Kerr. Strikingly, the absolute and relative penetration have *opposite* winners (Fig. 21): under evaporation the absolute periapsis descends but the horizon $2m(v)$ retreats faster, so the relative penetration $r_{\min}/2m(v_{\rm peri})$ *worsens* for later arrivals; under accretion the periapsis rises yet the horizon grows faster, so late arrivals graze the relative horizon. The orbit–horizon chase has reversed outcomes measured in r versus in horizon units.

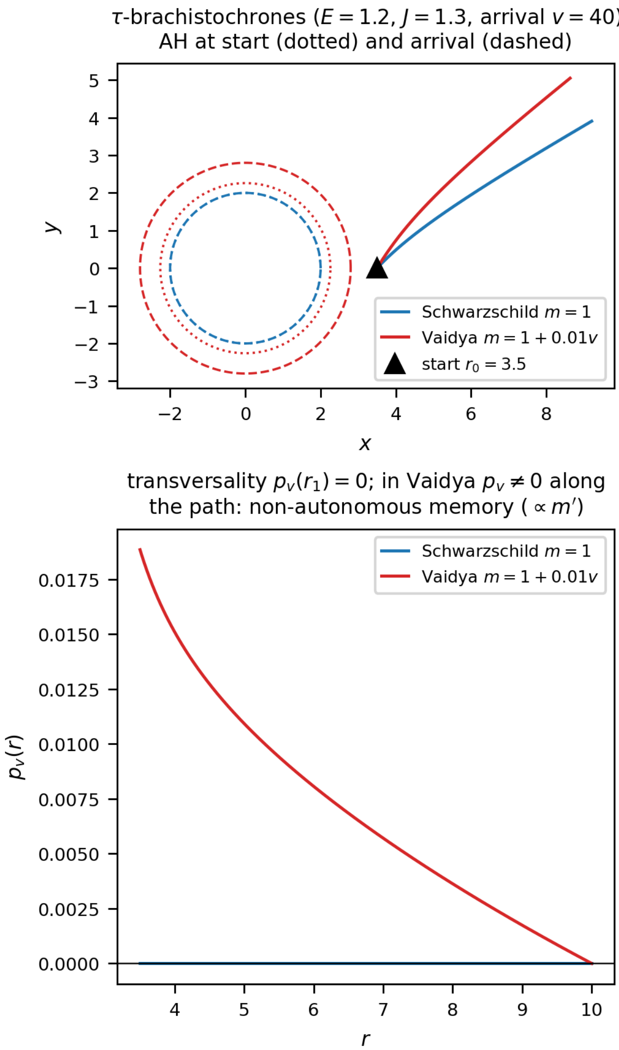

FIG. 5. τ-brachistochrones Schwarzschild vs Vaidya, and $p_v(r)$: transversality $p_v(r_1) = 0$, non-autonomous memory $\propto m'$.

C. Plunge-radius law and the (absent) evaporative inversion

The turning-point law has the static Schwarzschild form evaluated at the mass at periapsis, $f_* = 1 - 2m(v_{\rm peri})/r_{\min}$:

$$\tau: \quad (\hat{E}^2 - f_*)J^2 = f_* r_{\min}^2, \tag{13}$$

$$t: \quad (\hat{E}^2 - f_*)J^2 = \hat{E}^2 r_{\min}^2/f_*, \tag{14}$$

plus a genuine $O(m')$ correction from the teleological momentum p_v. The static ratio $J_t^2/J_\tau^2 = \hat{E}^2/f^2 > 1$ gives $r_{\min}^t < r_{\min}^\tau$ ("t deeper"), robust for physical evaporation: *there is no physical evaporative inversion* (the apparent linear-model inversion is an artifact of $m \leq 0$).

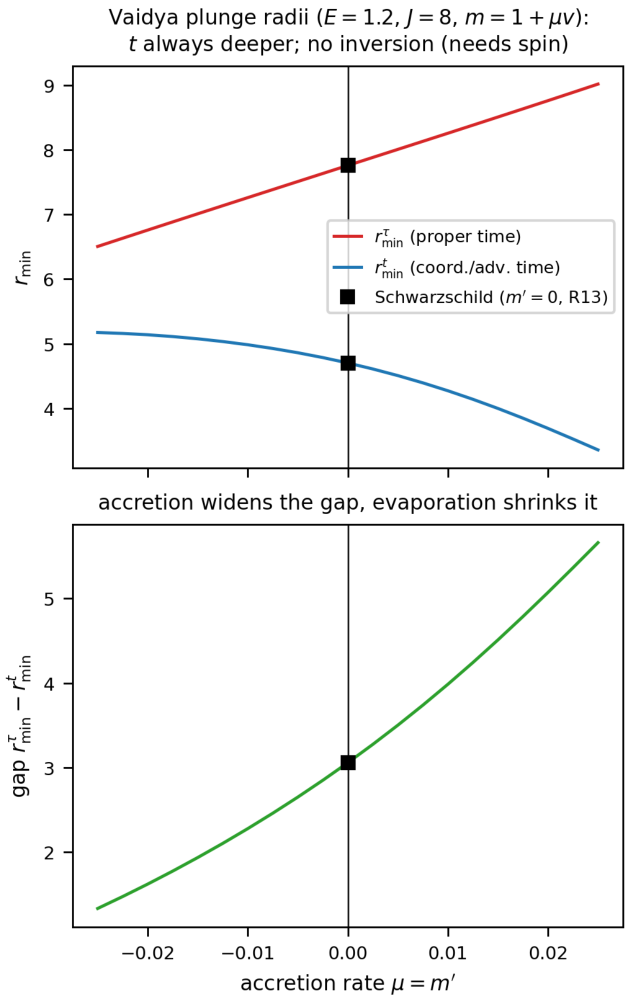

FIG. 6. Plunge radii $r_{\min}^{t,\tau}$ and gap vs $\mu = m'$; t deeper, accretion widens the gap.

V. THAKURTA–KERR: CONFORMAL ROTATING BLACK HOLE

Thakurta–Kerr is conformal Kerr, $g = A(\eta)^2 g_{\rm Kerr}$ [12, 25], one of a family of cosmological black holes [21, 26–30]. For constant A the transfer theorem maps the problem to Kerr with $E_{\rm eff} = \hat{E}/A$, $J_{\rm eff} = J/A$; the horizon r_+ is rigid (conformally invariant), the ergosphere $r_e = 2M$ is crossed regularly, while the freezing surface $A^2 f = \hat{E}^2$ breathes.

a. Indicatrix and Hamiltonians. Equatorially, write $\Delta = r^2 - 2Mr + s^2$, $\bar{v}^2 = 1 - A^2 f/\hat{E}^2$, and $\bar{P} =$

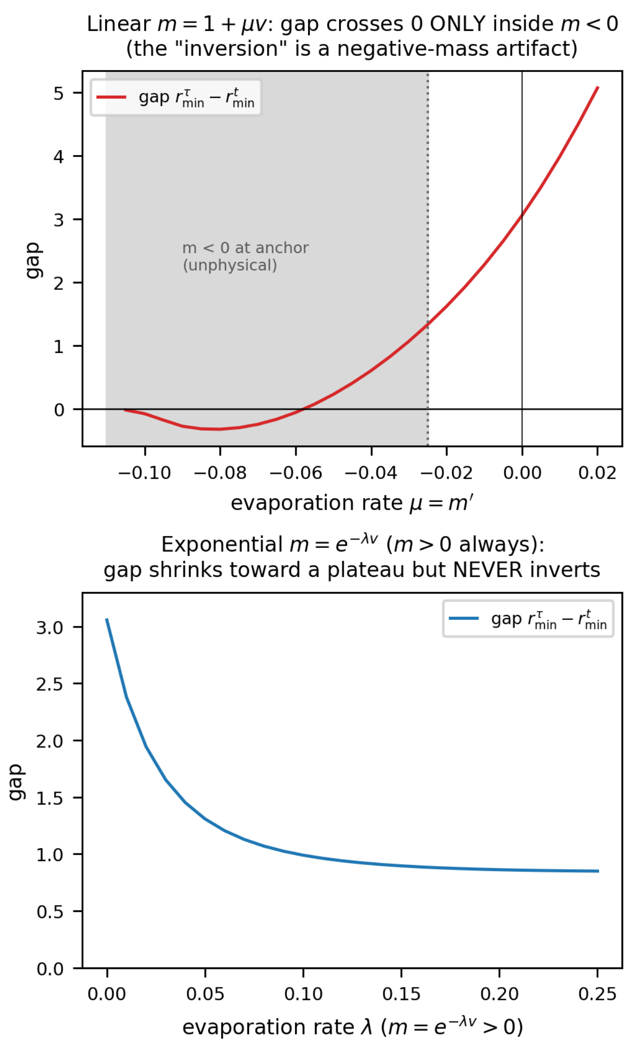

FIG. 7. No physical evaporative inversion: the linear-model sign change lies entirely in the unphysical $m < 0$ region; with $m = e^{-\lambda v} > 0$ the gap shrinks but never inverts.

$P + A^2(2Ms/r)^2/\hat{E}^2$ with $P = r^2 + s^2 + 2Ms^2/r$. The rail $\hat{E} = -u_\eta$ makes the indicatrix an ellipse with an *azimuthal* wind (frame dragging),

$$\varphi_0' = \frac{2Ms}{r}\frac{\bar{v}^2}{\bar{P}}, \qquad R^2 = f\bar{v}^2 + \bar{P}\varphi_0'^2 = \frac{\Delta\,\bar{v}^2}{\bar{P}}, \tag{15}$$

the last equality using the Kerr identity $fP + (2Ms/r)^2 = \Delta$. The conformal-time and proper-time Hamiltonians are

$$H_\eta = p_\varphi\varphi_0' + R\sqrt{(\Delta/r^2)p_r^2 + p_\varphi^2/\bar{P}} - 1, \tag{16}$$

$$H_\tau = \tilde{p}_\varphi\varphi_0' + R\sqrt{(\Delta/r^2)p_r^2 + \tilde{p}_\varphi^2/\bar{P}} - \frac{A^2 f}{\hat{E}}, \tag{17}$$

with the conformal gravitomagnetic shift $\tilde{p}_\varphi = p_\varphi - 2MsA^2/(r\hat{E})$. Both are regular through the ergosphere ($f = 0 \Rightarrow R^2 = \bar{v}^2\Delta/\bar{P} > 0$) and degenerate only at the horizon $\Delta = 0$, which is therefore conformally invariant—independent of A and $\hat{E}$. The three branches share this ellipse: t follows the same curves as η through $t = \int A\,\mathrm{d}\eta$, while τ is Riemannian-pure, the wind cancelling the gravitomagnetic term, so it survives the expansion (Fig. 8).

b. Non-autonomous optical reduction. Solving the indicatrix for $\mathrm{d}\eta$ at fixed spatial displacement casts the η-branch as a closed-form Randers metric, non-autonomous through the conformal factor,

$$\mathrm{d}\eta = n(\eta, r)\,\alpha_K + \beta_K, \qquad n = \frac{\hat{E}}{\sqrt{\hat{E}^2 - A^2 f}}, \tag{18}$$

$$\alpha_K^2 = \frac{r^2\,\mathrm{d}r^2}{f\Delta} + \frac{\Delta\,\mathrm{d}\varphi^2}{f^2}, \qquad \beta_K = -\frac{2Ms}{rf}\,\mathrm{d}\varphi. \tag{19}$$

The Riemannian data α_K and the one-form β_K are *exactly* those of the *null* Fermat (Randers) metric of Kerr [5]: all of the mass, rail, and expansion physics is carried by the scalar index n, which multiplies only the Riemannian part, while the frame-dragging one-form β_K is rigid—conformally invariant and independent of $\hat{E}$. The reduction is thus a Zermelo problem with a time-dependent landscape in which only the metric term breathes, not the wind. It degenerates at the ergosphere ($f = 0$, where $1/F$ is intrinsic, as for stationary Kerr) even though the Hamiltonian (16) remains regular there; the limits $\hat{E} \to \infty$ (null Kerr optics), $A = 1$ (stationary Perlick on Kerr), $s = 0$ (Thakurta–Schwarzschild) and $M = s = 0$ ($n = 1/v$, FLRW) are all recovered.

A. Quasi-constants and drift

For constant A the transfer theorem is exact at the level of conserved quantities: $u_{\mathrm{TK}} = u_{\mathrm{K}}/A$ maps the $\hat{E}$-rail onto the Kerr rail with $E_{\mathrm{eff}} = \hat{E}/A$, so every Kerr quasi-constant carries over by substitution,

$$K^{\mathrm{TK}} = K^{\mathrm{Kerr}}(E_{\mathrm{eff}}), \qquad K_\tau = Q_{\mathrm{std}} + a^2 f_2\,p_\theta^2 + O(a^4), \tag{20}$$

with Q_{std} the Carter constant and f_2 the next-to-leading polar coefficient [8, 31]. The polar libration that carries the quasi-constant lives off the equatorial plane, governed by the full Hamiltonian on a generic slice $\Sigma = r^2 + s^2\cos^2\theta$,

$$H_\tau = \tilde{p}_\varphi\varphi_0' + R\sqrt{\frac{\Delta}{\Sigma}p_r^2 + \frac{p_\theta^2}{\Sigma} + \frac{\tilde{p}_\varphi^2}{\bar{G}}} - \frac{A^2 f_\Sigma}{\hat{E}}, \tag{21}$$

with $f_\Sigma = 1 - 2Mr/\Sigma$, $b = 2Mar\sin^2\theta/\Sigma$, $\bar{G} = G + A^2b^2/\hat{E}^2$, $\tilde{p}_\varphi = J - A^2b/\hat{E}$, $\varphi_0' = b\bar{v}_\Sigma^2/\bar{G}$, $R^2 = \bar{v}_\Sigma^2\Delta\sin^2\theta/\bar{G}$, and the block identity $f_\Sigma G + b^2 = \Delta\sin^2\theta$; it reduces to Eqs. (16)–(17) on the equator. For $A = A(\eta)$ the substitution (20) is only the zeroth order: the transport equation $\mathrm{d}K/\mathrm{d}\lambda = \{K, H\} + \partial_\eta K$ acquires an ex-

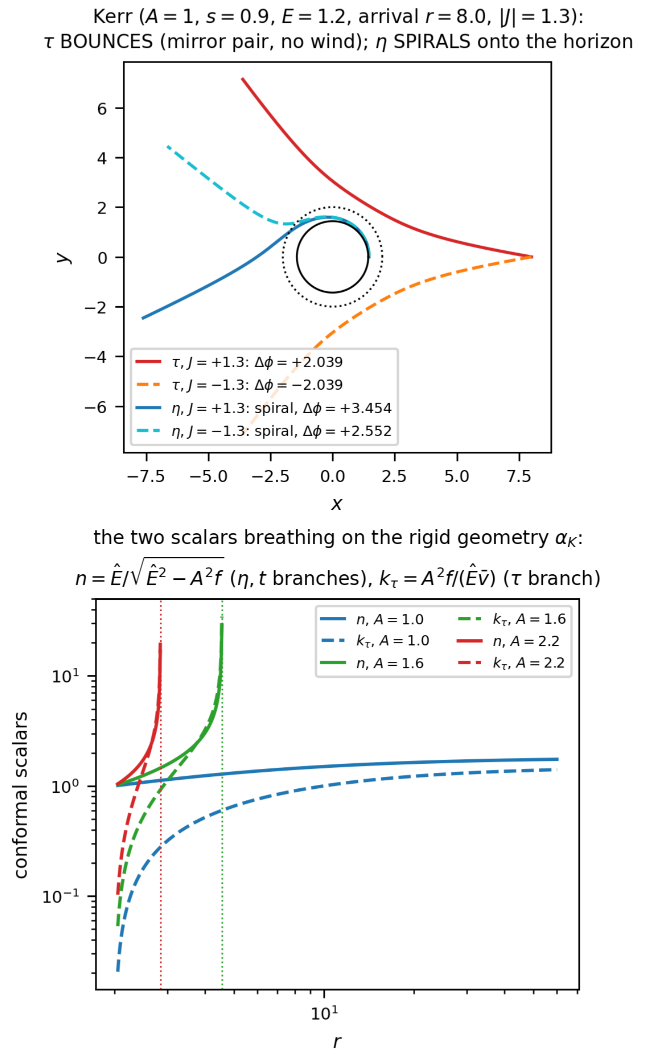

FIG. 8. The three branches: τ bounces (mirror pair, no wind), η spirals (gravitomagnetic wind); breathing conformal scalars n, k_τ.

plicit source at order $a^2\varepsilon$, $\varepsilon = A'/A$,

$$\left.\frac{\mathrm{d}K}{\mathrm{d}\eta}\right|_{\mathrm{expl}} = \varepsilon\, a^2\, S_1, \qquad S_1 = 2E_{\mathrm{eff}}^2\left[\cos^2\theta + \frac{M\cos 2\theta\, p_\theta^2}{r\, DE_0^2}\right], \tag{22}$$

where $DE_0 = r^2 w_{\mathrm{eff}}$. The counterterm h solving $\{h, H_0\} = -S_1$ inherits the double pole at the *freezing wall* $r_w = 2M/(1-E_{\mathrm{eff}}^2)$, so the dynamical quasi-constant is

$$K^{\mathrm{TK}}(\eta) = K^{\mathrm{Kerr}}\big(E_{\mathrm{eff}}(\eta)\big) - a^2 \int \varepsilon\, S_1\, \mathrm{d}\eta, \tag{23}$$

with closed toroidal averages $\langle S_1\rangle = E_{\mathrm{eff}}^2(1 - J^2/L^2) + 2ME_{\mathrm{eff}}^2|J|(L - |J|)/(r\, DE_0^2)$, $L^2 = p_\theta^2 + J^2/\sin^2\theta$. In Kerr ($E > 1$) the wall is absent; in Thakurta–Kerr with $E_{\mathrm{eff}} < 1$ it enters from the far field and advances inward as A grows, becoming a *global attractor*: the bound τ-libration ceases to exist, so the dynamical quasi-constants live on scattering arcs ($E_{\mathrm{eff}} > 1$) or on the t-branch.

VI. EQUATORIAL CLOSED FORMS AND THE ERGOSPHERE

A. Separatrix in Weierstrass functions

From the doranTau factorization $\Delta - J_c^2 w = f(r^2 + c^2)$, $c = a/E$ [8], the equatorial τ-separatrix $\varphi(r)$ is genus 1, closed in the Weierstrass functions $\wp, \sigma, \zeta$ [32–35]: the angle is linear in the uniformizing variable $z(r) = \wp^{-1}(A/r+B)$ plus two σ-ratios [App. A 1, Eq. (A3)], regular *through* the ergosphere because the factor $r^2 + c^2$ divides the turning quartic and removes r_e from the radicand. The Boyer–Lindquist form winds logarithmically at the horizon; the Doran shift—a second, quasi-lemniscatic elliptic curve—cancels it, so the ergosphere-penetrating separatrix $\varphi_D = \varphi_{\mathrm{BL}} + \varphi_{\mathrm{shift}}$ is closed on two elliptic curves. The Kerr geodesic structure and its constants are classical [31, 36]. Validated end-to-end to 2.8×10^{-10} through the ergosphere, and general across parameters.

B. Trichotomy and the ergosphere cusp

For the equatorial τ-branch, $\mathrm{d}\varphi/\mathrm{d}r \sim K\sqrt{r - r_e}$ as $r \to r_e$ (because $\sqrt{f} \to 0$ while the turning radicand stays finite), giving

$$\varphi - \varphi_e \sim \tfrac{2}{3}K\,(r - r_e)^{3/2}, \tag{24}$$

a cusp with radial tangent. The trichotomy in $|J|$ (sign of $a^2 - J^2E^2$): smooth periapsis above r_e ($|J| > J_c$); smooth crossing at r_e ($|J| = J_c = a/E$, separatrix); cusp reflection ($|J| < J_c$). The crossing branch is continued through r_e to the horizon in Doran (horizon-penetrating) coordinates [37–39], $\varphi_D = \varphi_{\mathrm{BL}} - \varphi_{\mathrm{shift}}$.

a. Retrograde branch and the τ/t asymmetry. The trichotomy above is a τ-branch statement, and it is *symmetric* in the sign of J: because the doranTau factorization depends on J only through J^2, the τ-capture band—the momenta for which the ingoing orbit reaches r_e—is $|J| \le J_c = a/E$, prograde and retrograde sharing the same threshold despite frame dragging. The t-branch is not protected. Its turning function (from $H_\eta = 0$ at $p_r = 0$, with $b = 2Ma/r$) is

$$N(r) = \left(\bar{P} - Jb\bar{v}^2\right)^2 - J^2\Delta\bar{v}^2, \tag{25}$$

which is only linear-plus-quadratic in J. At the ergosphere $N(r_e) = \bar{P}_e\,(\bar{P}_e - 2aJ)$ with $\bar{P}_e = \bar{P}(r_e) = 4M^2 + 2a^2 + a^2/E^2$, so the t-orbit crosses r_e at the single *prograde* value

$$J_+^t = \frac{\bar{P}_e}{2a} = \frac{4M^2 + 2a^2 + a^2/E^2}{2a}; \tag{26}$$

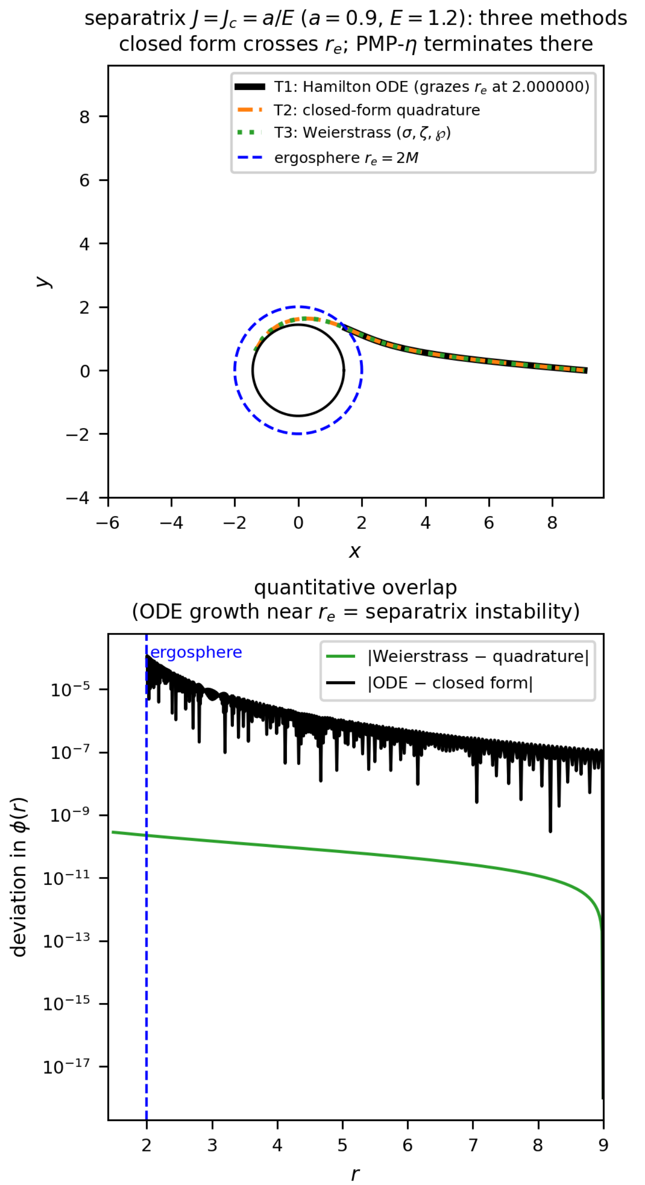

FIG. 9. Separatrix $J = J_c$: ODE, quadrature and evaluated Weierstrass agree to 2.8×10^{-10}; the PMP-η terminates at r_e.

there is no retrograde r_e-crossing. Instead the retrograde t-capture is bounded by the *marginal* (double-root) condition $N = N' = 0$ at a radius $r_* > r_e$—the massive analogue of the retrograde photon sphere—which closes to

$$J_c^- = -\left.\frac{\bar{P}}{\bar{v}\,(\sqrt{\Delta} - b\bar{v})}\right|_{r_*} < 0. \tag{27}$$

For $a = 0.9$, $E = 1.2$ (and $A = 1$) these give $J_+^t = 3.43$ and $J_c^- = -8.05$ ($r_* = 3.51$): the t-branch captures the wide, asymmetric band $J \in [J_c^-, J_+^t]$, far more capture-prone on the retrograde side. A moderately retrograde orbit ($J = -5$, say) is therefore captured on the t-branch but scattered on the τ-branch. For constant A the results carry over by $E \to \hat{E}/A$, $a \to s$, $J \to J/A$; in particular $J_c^\tau = sA^2/\hat{E}$.

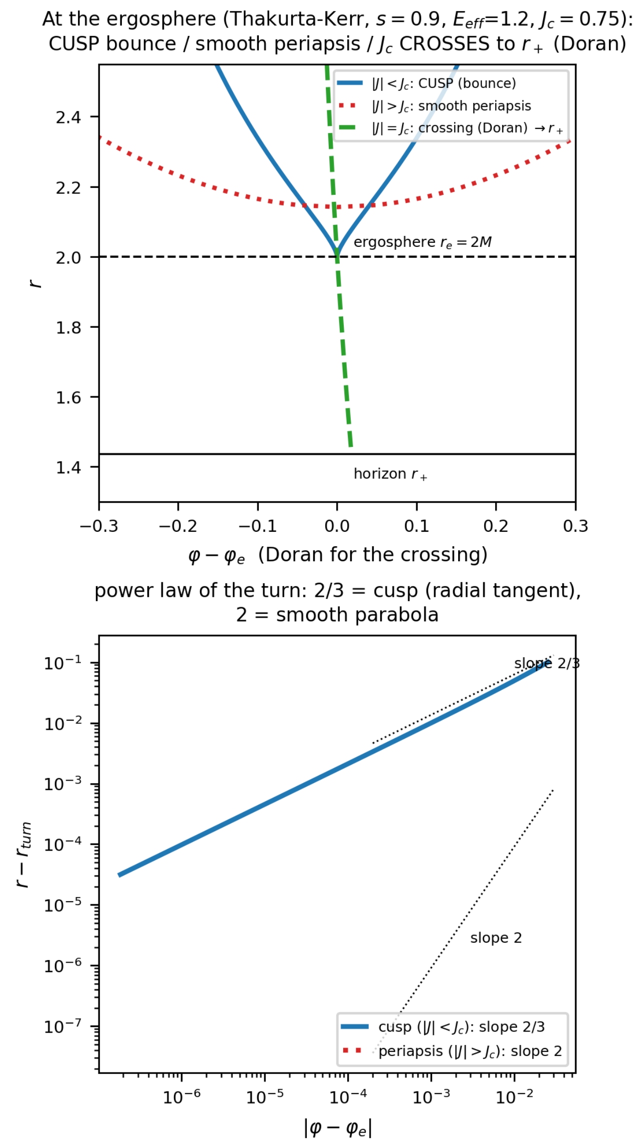

FIG. 10. Reflection at the ergosphere: cusp ($|J| < J_c$), smooth crossing to r_+ in Doran coordinates ($|J| = J_c$), smooth periapsis ($|J| > J_c$); power law 2/3 vs 2.

C. Plunge inversion: rotational and conformal

Two mechanisms invert the t/τ ordering. Rotationally, in the (a, J) plane, $\rho(r_*)^2 = H_{t0}/H_{\tau 0}$, extending the stationary Kerr analysis [8]. Conformally, the inversion locus is

$$\mathrm{cond}(r, A) = A^2 b\, Q - \bar{P}(\hat{E} - A^2 f) = 0, \quad Q = b\bar{v}^2 + \sqrt{\bar{v}^2 \Delta}, \tag{28}$$

with $\mathrm{cond}(1) < 0$ and $\mathrm{cond}(A_{\mathrm{freeze}}) = \hat{E}r\Delta(\hat{E}-1)/(r-2M) > 0$, so by the intermediate value theorem $A_{\mathrm{inv}} \in (1, A_{\mathrm{freeze}})$: the conformal inversion always exists and is reachable (far field $A_{\mathrm{inv}} \to \sqrt{\hat{E}}$).

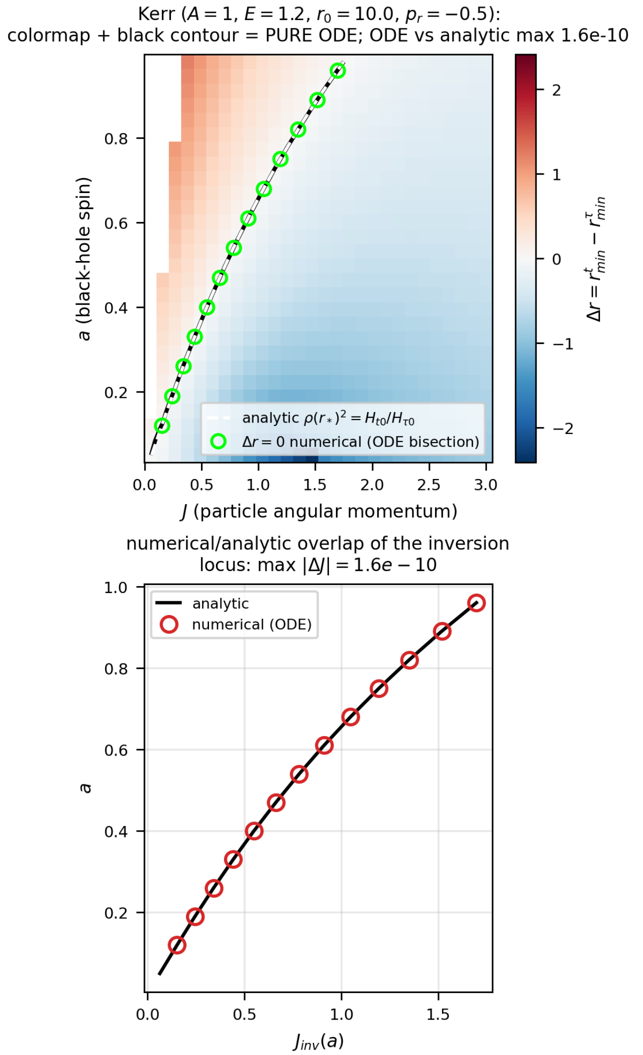

FIG. 11. Rotational inversion in the (a, J) plane; ODE locus vs analytic $\rho(r_*)^2 = H_{t0}/H_{\tau 0}$.

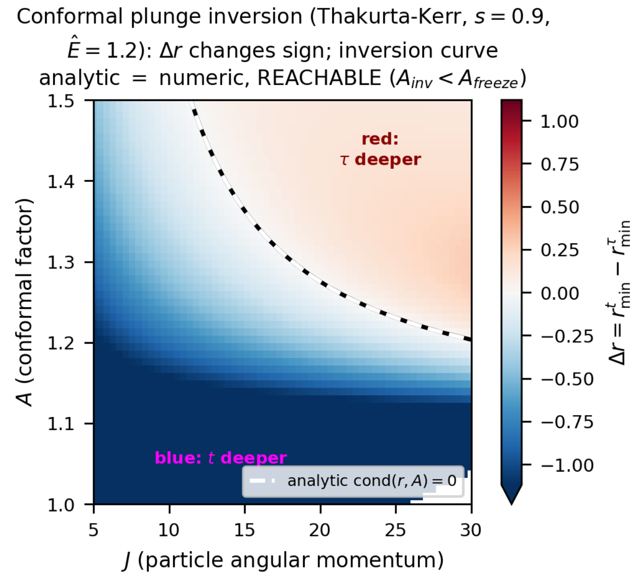

FIG. 12. Conformal inversion in the (J, A) plane: Δr changes sign; analytic cond $= 0$ matches the numerical contour; reachable ($A_{\mathrm{inv}} < A_{\mathrm{freeze}}$).

TABLE I. Protocol dependence of plunge inversion. Same-launch comparison inverts under either rotation or a conformal factor; the genuine fixed-endpoint brachistochrone never inverts.

protocol	spin a	conformal A
same launch	inverts (§VI C)	inverts (§VI C)
fixed endpoints	no ($\Delta r > 0$)	no ($n_t/n_\tau = E/f > 1$)

D. Fixed-endpoint comparison and protocol dependence

The statement "branch X plunges deeper" is protocol dependent. With *fixed initial data* (same r_0, J, p_r) the t-branch is deeper ($r^t_{\mathrm{min}} < r^\tau_{\mathrm{min}}$). With *fixed endpoints* (the genuine two-point brachistochrone), the two branches join the same A, B but are distinct paths, and the ordering reverses: $r^\tau_{\mathrm{min}} < r^t_{\mathrm{min}}$. The optical-index ratio $n_t/n_\tau = E/f > 1$ penalizes deep radii more heavily for the t-branch, so the t-brachistochrone bows outward while the τ-branch dips. This must be specified when comparing depths.

Crucially, neither driver of same-launch inversion survives the fixed-endpoint constraint. Scanning the black-hole spin over the entire Kerr flow with fixed symmetric endpoints $(r_0, \pm\Phi)$ leaves $\Delta r = r^t_{\mathrm{min}} - r^\tau_{\mathrm{min}}$ strictly positive ($\Delta r \in [+0.45, +2.06]$ on $a \in [0.05, 0.95]$); the map is nearly flat in a, its gradient set almost entirely by the endpoint angle Φ (encoded by J). The conformal scan (via the transfer $E_{\mathrm{eff}} = \hat{E}/A$) is likewise everywhere positive ($\Delta r \in [+0.002, +3.711]$), as forced by the theorem $n_t/n_\tau = E/f > 1$. Table I summarizes the resulting protocol dependence: inversion is a property of the *comparison protocol*, not an invariant of the two brachistochrones—it requires the freedom to let the endpoint move.

The verdict is independent of the endpoint symmetry: repeating both scans with *asymmetric* fixed endpoints ($r_A = 10, r_B = 6$) leaves $\Delta r > 0$ everywhere as well (spin $[+0.76, +2.64]$, conformal $[+0.16, +3.24]$; Figs. 16–17), consistent with the pointwise bound $n_t/n_\tau = E/f$.

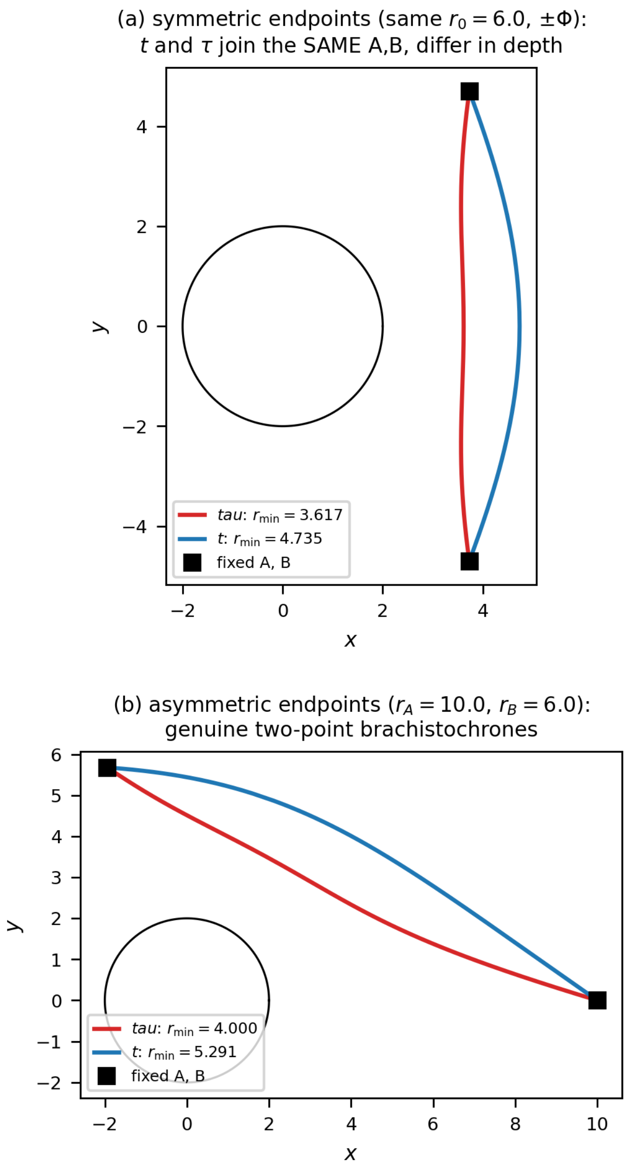

FIG. 13. Fixed-endpoint t and τ brachistochrones (Schwarzschild, $E = 1.2$): symmetric (top) and asymmetric (bottom) endpoints. Both branches join the same A, B; the τ-branch dips deeper.

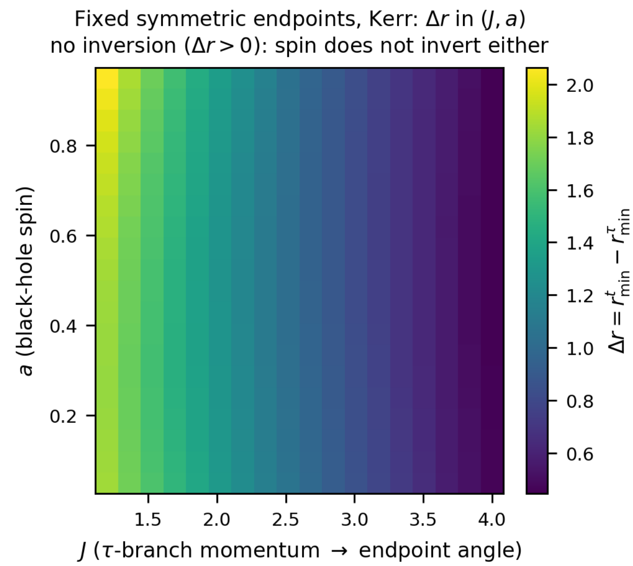

FIG. 14. Spin scan at fixed symmetric endpoints (Kerr, Hamiltonian flow): Δr in the (J, a) plane stays positive everywhere—spin does not invert once the endpoints are pinned. The gradient is set by the endpoint angle (J), not by a.

VII. CONCLUSIONS

The non-stationary brachistochrone is a well-posed controlled-rail problem: the invariant $-u \cdot W = \hat{E}$ is legitimate via Pontryagin's principle, and the selector W is fixed canonically by the Killing–CKV–Kodama hierarchy, with the Kodama energy conserved along the rail even when free fall does not conserve it. FLRW is the degenerate base—freezing without branch splitting—while the black hole breaks the t/τ symmetry through the gravitational gradient. On the equatorial sector we obtained closed-form separatrices in Weierstrass functions, continued through the ergosphere in Doran coordinates, and an analytic cusp theorem for the $|J| < J_c$ reflection. The plunge inversion resolves into a three-way picture: rotation and the conformal factor invert the t/τ ordering under a same-launch comparison, whereas physical evaporation never does, and *no* inversion survives once both endpoints are fixed. Outlook: the exact Sultana–Dyer source [27], the McVittie/Schwarzschild–de Sitter limits [26, 28, 40] with their photon surfaces [41], escape maps, and freezing-surface dynamics.

Appendix A: Validation and consistency checks

We collect the numerical checks that underpin the analytic results. The minimum principle is verified directly by perturbing the computed brachistochrones and confirming that both T_t and T_τ are minimized (Fig. 3 for the FLRW branch degeneracy, and Fig. 19 for the two distinct Vaidya branches). Stationary-limit residuals confirm that the non-stationary Hamiltonian flow reproduces the closed-form Kerr expressions as $m' \to 0$ and $A \to 1$ (Fig. 18); the Kodama energy is conserved to $\sim 10^{-16}$ along the rail (Fig. 4). The equatorial closed forms are validated end-to-end by the three-method superposition (ODE flow, quadrature, evaluated Weierstrass) through the ergosphere (Fig. 9), and the fixed-endpoint no-inversion theorem is checked over the asymmetric-endpoint grids (Figs. 16–17).

1. Explicit closed forms

a. Equatorial separatrix in Weierstrass functions. On the separatrix $J = J_c$ the doranTau factorization $\Delta - J_c^2 w = f(r^2 + c^2)$, $c = a/E$, makes the turning quartic $Q_4 = r\big((E^2-1)r + 2M\big)(r^2 + c^2)$, and the factor $r^2 + c^2$ divides Q_4 so that only the horizon poles survive. Writing

$$\begin{aligned} A &= \frac{Ma^2}{2E^2}, \qquad B = \frac{a^2(E^2-1)}{12E^2}, \\ g_2 &= \frac{a^4(E^2-1)^2}{12E^4} - \frac{a^2M^2}{E^2}, \\ g_3 &= \frac{a^4\big[36E^2M^2(1-E^2) - a^2(E^2-1)^3\big]}{216E^6}, \end{aligned} \tag{A1}$$

the uniformizing variable is $z(r) = \wp^{-1}\big(A/r + B; g_2, g_3\big) = \int_0^r \mathrm{d}r'/\sqrt{Q_4}$, the Abel map of the turning quartic Q_4: it is the elliptic argument in which the radial motion linearizes, so that r is recovered by $A/r + B = \wp(z)$ and the physical range $r \in [r_{\min}, \infty)$ maps to a segment of z. With $r_\pm = M \pm \sqrt{M^2 - a^2}$ the horizons,

$$\begin{aligned} \alpha_\pm &= \frac{r_\pm\big((E^2-1)r_\pm + 2M\big)}{r_\pm - r_\mp}, \\ \lambda_\pm &= \frac{\alpha_\pm}{\sqrt{Q_4(r_\pm)}}, \qquad v_\pm = z(r_\pm), \end{aligned} \tag{A2}$$

$\wp(v_\pm) = A/r_\pm + B$, and $\Lambda_0 = (E^2-1) - \alpha_+/r_+ - \alpha_-/r_- + 2\lambda_+\zeta(v_+) + 2\lambda_-\zeta(v_-)$, the angle is linear in the uniformizing variable $z \equiv z(r)$ plus two Weierstrass-σ ratios,

$$\begin{aligned} \varphi(r) = \frac{a}{E}\Big[&\Lambda_0\, z(r) + \lambda_+ \ln\frac{\sigma(z - v_+)}{\sigma(z + v_+)} \\ &+ \lambda_- \ln\frac{\sigma(z - v_-)}{\sigma(z + v_-)}\Big] + \text{const.} \end{aligned} \tag{A3}$$

Because $Q_4(2M) = 4M^2(4E^2M^2 + a^2) > 0$, Eq. (A3) is regular *through* the ergosphere $r_e = 2M$; the only singularities are the third-kind logarithms at the horizons. The conformal case is $E \to E_{\text{eff}} = \hat{E}/A$ in every parameter.

b. Doran continuation through the horizon. The Boyer–Lindquist form $\varphi(r)$ winds logarithmically at r_+ (the standard BL artifact); adding the Doran shift $\mathrm{d}\varphi_{\text{shift}} = 2Mar\,\mathrm{d}r/(\Delta\sqrt{C_3})$ with the cubic $C_3 = 2Mr(r^2 + a^2)$ regularizes it. This is a *second* elliptic curve, in clean Weierstrass form $r(z) = (2/M)\wp(z; g_2, g_3)$ with $g_2 = -M^2a^2$, $g_3 = 0$ (quasi-lemniscatic), giving $\varphi_{\text{shift}} = \frac{a}{r_+ - r_-}\big(B_+ - B_-\big)$, $B_\pm = 2\zeta(v_\pm)z + \ln[\sigma(z - v_\pm)/\sigma(z + v_\pm)]$; the log divergences of φ and φ_{shift} cancel at r_+, the analytic content of Doran regularity. Thus $\varphi_D = \varphi_{\text{BL}} + \varphi_{\text{shift}}$ is closed on two elliptic curves.

c. Stationary-limit conserved quantities. At $A = 1$ the branch flows admit the closed first integrals used for validation,

$$\mathcal{K}_\tau = J, \qquad \mathcal{K}_t = \frac{fJ + 2Ms/r}{E}, \qquad \mathcal{K}_\eta = \frac{fJ}{E}, \tag{A4}$$

the equatorial Kerr doranTau/doranT constants.

d. Rotational inversion function. The same-launch t/τ ordering inverts where $\rho(r_*)^2 = H_{t0}/H_{\tau 0}$ with the explicit ratio of centrifugal potentials

$$\rho = \frac{A^2}{\hat{E}}\Big(f + \frac{2Ms}{rJ}\Big), \tag{A5}$$

$\rho < 1$ giving t deeper, $\rho > 1$ giving τ deeper.

e. Conformal trichotomy. With constant A the separatrix momentum is $J_c = sA^2/\hat{E}$, and above it the smooth periapsis is the outer root of

$$\Delta(r) - \Big(\tfrac{J}{A}\Big)^2\Big[\big(\tfrac{\hat{E}}{A}\big)^2 - f(r)\Big] = 0, \tag{A6}$$

while $|J| < J_c$ reflects at the cusp $r_{\min} = r_e$ and $|J| = J_c$ crosses to r_+.

f. Closed toroidal averages. The secular source $\langle S_1\rangle$ of Eq. (22) follows from the orbit-plane averages ($\cos\theta = \sin i\,\sin\psi$, $k = 1 - J^2/L^2$)

$$\Big\langle \cos^2\theta \Big\rangle = \tfrac{k}{2}, \quad \Big\langle \tfrac{1}{\sin^2\theta} \Big\rangle = \tfrac{L}{|J|}, \quad \big\langle p_\theta^2 \big\rangle = L(L - |J|), \tag{A7}$$

with $\langle \cos 2\theta\, p_\theta^2\rangle = |J|(L - |J|)$.

2. Consistency residuals

The closed forms and reductions above were checked symbolically and numerically. The Weierstrass separatrix (A3) reproduces the direct quadrature to $|\Delta\varphi| < 2.8 \times 10^{-10}$ through the ergosphere, with the $\wp$-identity satisfied to 9×10^{-28} and the antiderivative to 30 digits; the Doran shift reduces to 1.1×10^{-16}. The non-autonomous Randers data (19) match the Kerr null Fermat metric to $\sim 10^{-14}$. The equatorial Thakurta–Kerr flow reproduces the $A = 1$ Kerr constants (A4) to ten digits at $r = 4, 6, 9$ ($a = 0.9$, $E = 1.2$), and the $s \to 0$

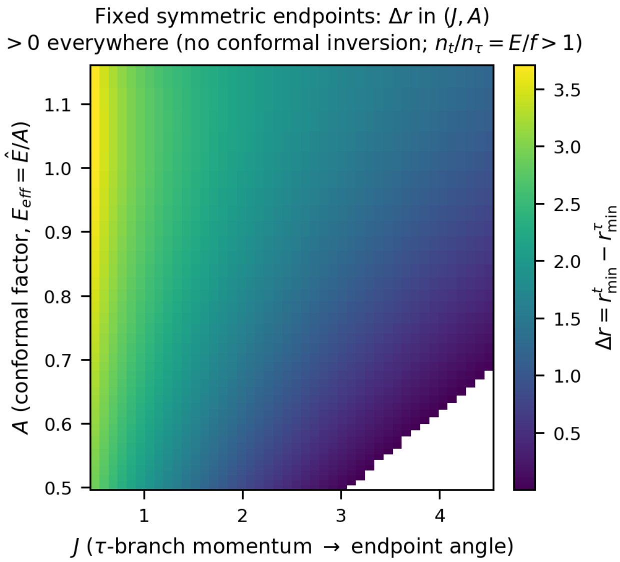

FIG. 15. Conformal scan at fixed symmetric endpoints (via $E_{\rm eff} = \hat{E}/A$): Δr in the (J, A) plane is positive everywhere, matching the closed-form theorem $n_t/n_\tau = E/f > 1$.

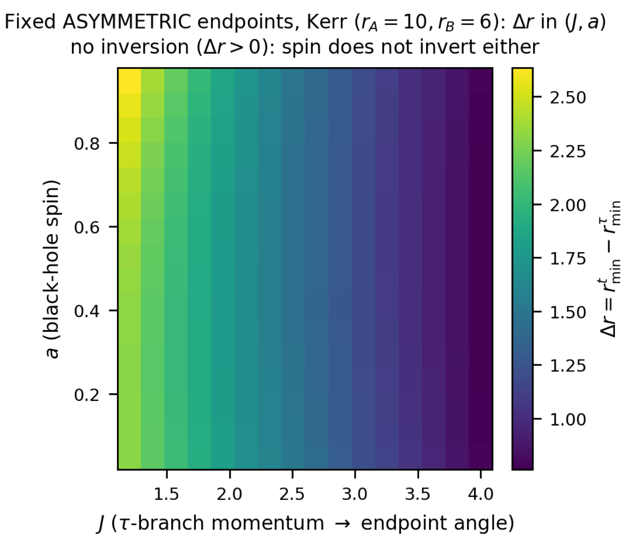

FIG. 16. Robustness of the fixed-endpoint no-inversion result under *asymmetric* endpoints ($r_A = 10, r_B = 6$), spin scan (Kerr flow): $\Delta r > 0$ throughout the (J, a) plane.

limit the Thakurta–Schwarzschild flow to 10^{-12}; the 3+1 Hamiltonian (21) reduces to the equatorial one exactly, and conserves L^2 to 6×10^{-13} at $a = 0$. The Kodama energy is held to $|-u \cdot K - \hat{E}| < 7 \times 10^{-16}$ along the rail, and the static inversion/trichotomy colormap agrees with the analytic contours (A6) to a relative 7×10^{-12}.

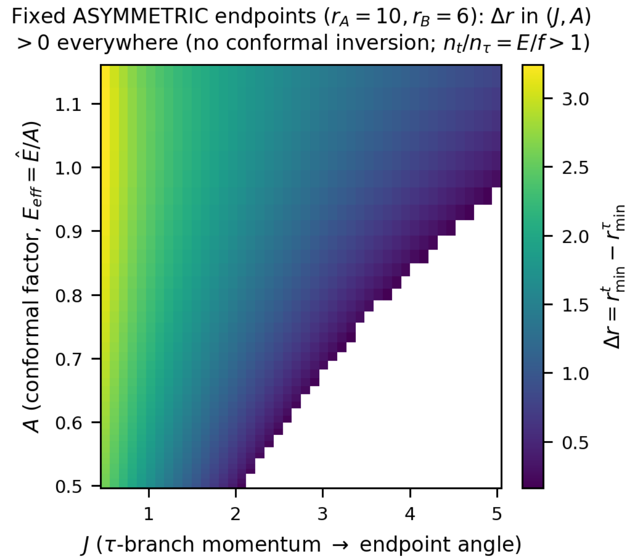

FIG. 17. Same robustness check, conformal scan (via $E_{\rm eff} = \hat{E}/A$): $\Delta r > 0$ throughout the (J, A) plane for asymmetric endpoints. The blank wedge is the region where the τ-turning point rises above r_B (no dipping orbit reaches B).

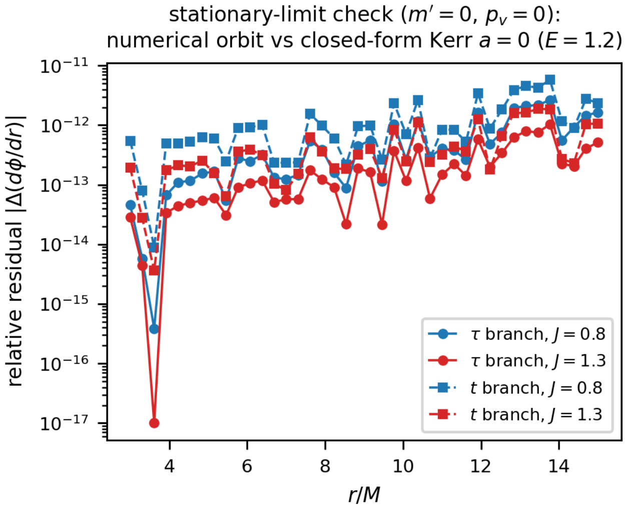

FIG. 18. Stationary-limit check ($m' = 0$): numerical orbit vs closed-form Kerr $a = 0$.

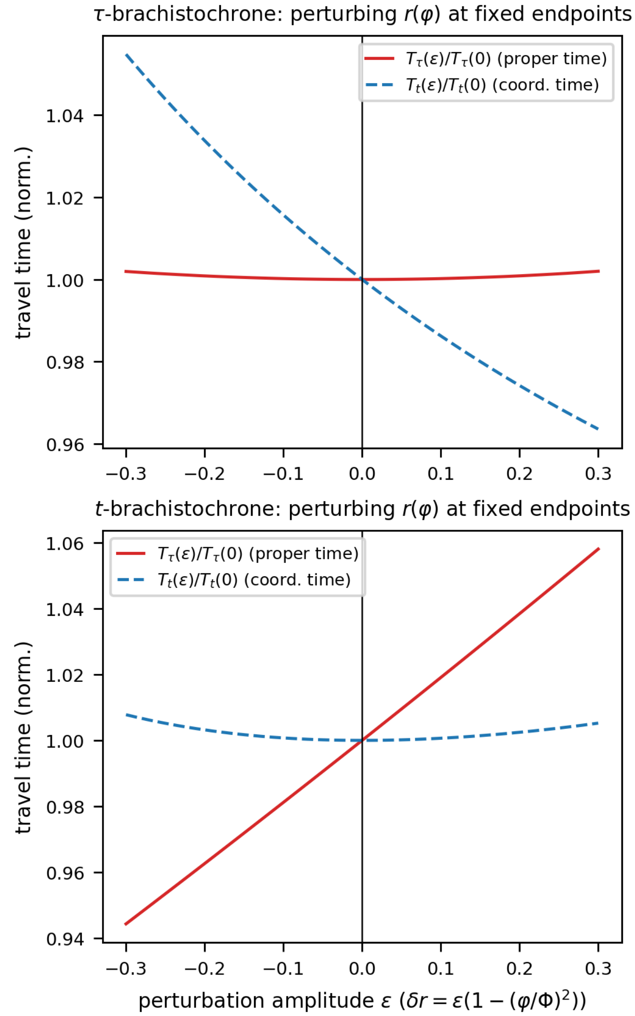

FIG. 19. Each branch minimizes its own travel time; t and τ are distinct curves.

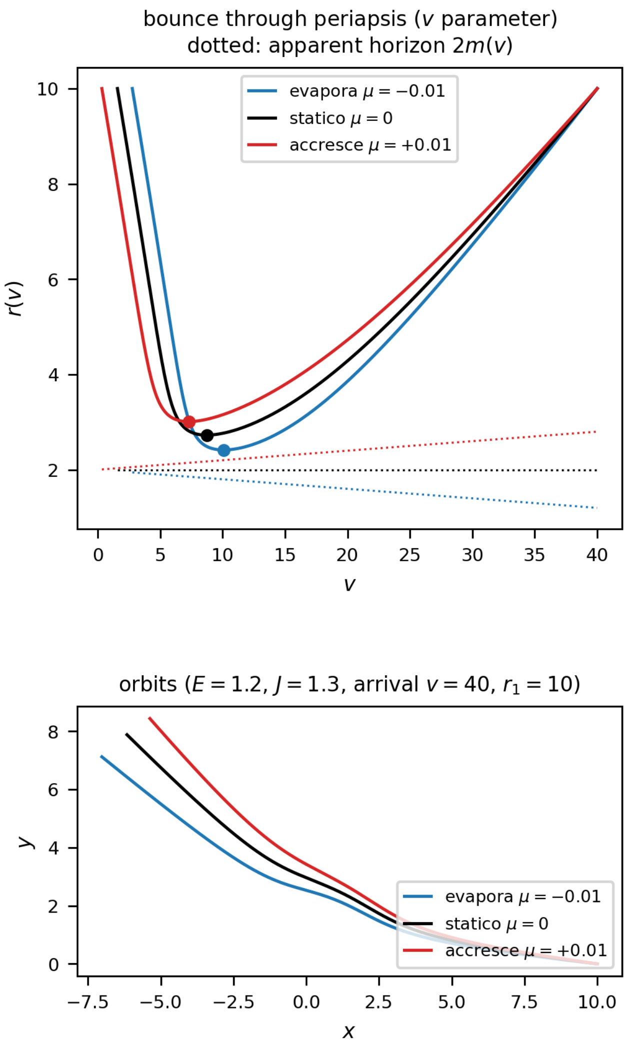

FIG. 20. Bounce through periapsis in the v-parametrization: the Euler–Lagrange flow is regular at $r' = 0$ and integrates through the turning point, reaching $r_{\min}$ exactly where the r-parametrization diverges.

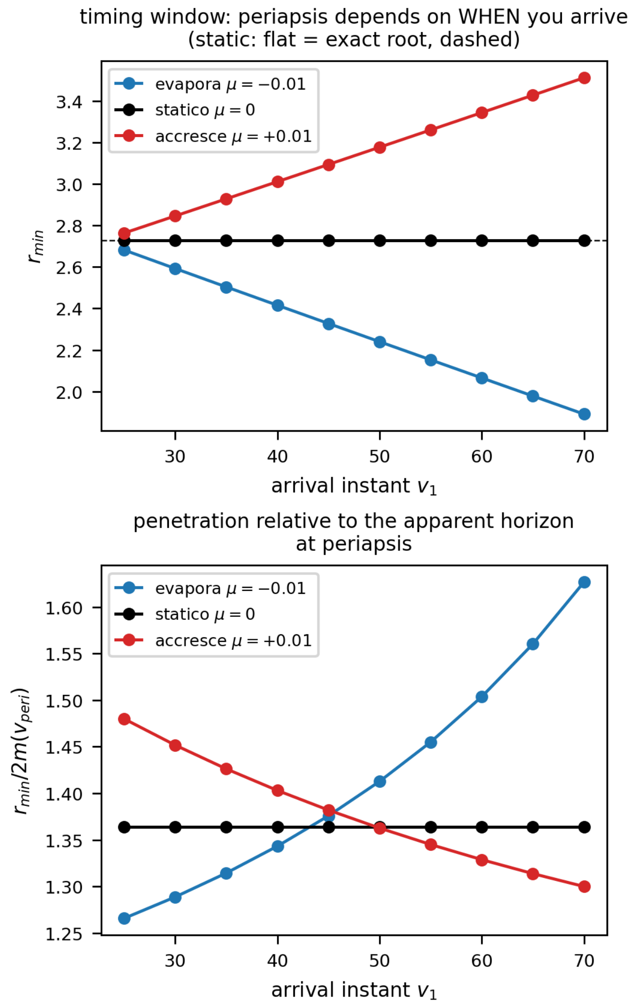

FIG. 21. Timing window: $r_{\min}$ depends on the arrival epoch v_1. The orbit–horizon race has opposite winners in absolute ($r_{\min}$) and relative ($r_{\min}/2m$) terms for accretion vs evaporation.

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DATA AVAILABILITY

The complete numerical and symbolic codebase underlying this work is openly available under the MIT license in the GitHub repository "NonStationaryBrachistochrone" [42]:

https://github.com/ImanetorX98/NonStationaryBrachistochrone

The repository includes the trajectory integrators, the Pontryagin Hamiltonian constructions, the Poisson-bracket closure and quasi-invariant scans (Sec. V A), the Weierstrass separatrix and Doran-continuation scripts, the fixed-endpoint BVP and colormap pipelines, and the figure-generation code. It also retains exploratory scripts and negative results, documenting which ansatz classes failed the held-out trajectory and phase-space residual tests reported in Sec. V A and App. A.

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